MILITARY SPECIFICATION

ANTENNA COUPLER GROUP, CU-1382( )
(HF BROADBAND, WIDE DYNAMIC RANGE)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the design and performance of two types of HF broadband, wide dynamic range solid state multicouplers. The two equipments described herein are for use at Naval communication facilities and shall provide coupling of from one to eight radio receivers to one antenna in a 50-ohm system.

1.2 Classification. Multicouplers covered by this specification shall be of the following types as specified:

Type I - High linearity
Type II - High linearity; phase correlated

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-E-16400	- Electronic, Interior Communication and Navigation Equipment, Naval Ship and Shore, General Specification for
MIL-E-17555	- Electronic and Electrical Equipment; Accessories and Repair Parts, Packaging and Packing of
MIL-F-19207	- Fuseholders, Extractor Post Type, Blown Fuse Indicating and Nonindicating, General Specification for
MIL-S-19500	- Semiconductor Devices, General Specification for

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes

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MIL-STD-108	- Definitions of and Basic Requirements for Enclosure for Electrical and Electronic Equipment
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-189	- Racks, Electrical Equipment, 19-Inch and Associated Panels
MIL-STD-242	- Electronic Equipment Parts, Selected Standards
MIL-STD-454	- Standard General Requirements for Electronic Equipment
MIL-STD-461	- Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of
MIL-STD-701	- List of Standard Semiconductor Devices
MIL-STD-781	- Reliability Tests: Exponential Distribution
MIL-STD-810	- Environmental Test Methods
MIL-STD-1631	- Procedure for Selection of Electronic and Electrical Parts During Equipment Design
MS 3057	- Clamp, Cable, Electrical Connector
MS 3452	- Connector, Receptacle, Electric, Box Mounting
MS 3456	- Connector, Plug, Electric, Rear Release, Crimp Contact, AN Type

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article sample. Prior to beginning production, a sample shall be tested as specified in 4.3 (see 6.3).

3.2 General. The equipment shall be in accordance with the provisions of MIL-E-16400 to the extent specified herein. Types I and II multicouplers shall be designed for Naval shore fixed plant use and shall provide optimum coupling between a single antenna and eight receivers in communication systems. The multicouplers shall be complete, including power supply in one unit, and shall contain all necessary parts and circuitry to provide the specified performance when connected to the specified source. Those requirements specified herein which do not refer to a specific type of multicoupler shall apply to both types. Those requirements which refer to a specific type of multicoupler shall apply to only that type.

3.2.1 Equipment composition. The multicoupler shall contain all parts, including semiconductor devices, as necessary to its performance, in accordance with the requirements of this specification, when connected to the primary power

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sources specified in 3.5.6. Connector fittings required for external cables shall be furnished with each equipment.

3.2.2 Design and construction. The design and construction of the equipment shall be predicated upon the use of modular assemblies conforming to the requirements of MIL-E-16400. Modular assemblies may be of the throw-away type of construction if the predicted mean-time-between-failure (MTBF) exceeds 100,000 hours (see 3.3.1). Those modules not meeting this requirement shall be of such design and construction as to be repairable in the field. The number of modular assemblies shall provide for not less than the following functional subdivisions:

(a) Input filter
(b) Amplifier/output isolator
(c) Power supply

3.2.3 Duty cycle. The equipment and all items entering into its construction shall be designed for both intermittent and continuous operation, in conformance with the performance requirements of this specification, under adverse conditions normally encountered in the Naval communications services ashore.

3.2.4 Material. Material entering into the manufacture of the equipment shall be in accordance with MIL-E-16400, except as otherwise specified herein. These materials shall have magnetic characteristic in accordance with MIL-E-16400. Particular attention shall be given to the acceptable and unacceptable material requirements of MIL-E-16400.

3.2.5 Selection of parts. The selection and application of all parts shall be in accordance with MIL-E-16400. MIL-STD-242 shall be the basic source for the selection of parts.

3.2.6 Procedure for selection of electronic and electrical parts and material during design of military items. Prescreening of non-standard parts approval requests shall be accomplished in accordance with MIL-STD-1631.

3.2.7 Tropicalization. Tropicalization of the complete equipment or any unit or part thereof will not be required.

3.2.8 Semiconductor devices (transistors, diodes). Semiconductor devices shall be selected from approved types in accordance with MIL-STD-701 and MIL-S-19500, as required by MIL-E-16400. Plastic encapsulated semiconductors shall not be used.

3.2.9 Printed wiring. Printed wiring, where employed, shall conform to Requirement 17 of MIL-STD-454.

3.2.10 Fuses. Each side of the primary power input to the power supply shall be separately fused. The fuses shall be of the miniature type and shall be mounted in insulated, extractor type fuseholders mounted on the front operating panel. The active fuseholders along with a spare fuseholder equipped with a spare fuse and spare blown fuse indicator shall be located on the front panel of each equipment.

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3.2.11 Blown fuse indicators. Blown fuse indicators, conforming to the requirements of MIL-F-19207, shall be provided for each active fuse.

3.2.12 Enclosure. The equipment housing shall conform to the requirements of MIL-STD-108 for a totally enclosed enclosure.

3.2.13 External finish. The external finish shall be in accordance with MIL-E-16400.

3.2.14 Control knobs. Control knobs shall conform to the requirements of MIL-E-16400. Tactile identification forms of control knobs will not be required.

3.2.15 Input and output connectors. RF input and output connectors shall be mounted at the rear of the unit and shall be of the male BNC type. Connectors shall be spaced sufficiently to permit adapters for the type N connectors to be used. A mating connector for each BNC input and output connector shall be furnished with the equipments. The manufacturer shall inquire of the procuring activity as to what type of coaxial cable is to be used with these equipments, and on the basis of this information furnish mating connectors which properly interface with the designated type coaxial cable.

3.2.16 Mounting. The complete multicoupler equipment shall be designed such that two can be fitted into a 1.719-inch (43.66 mm) by 19-inch (48.26 cm) relay rack which conforms with MIL-STD-189. The equipment chassis shall have a maximum depth of 18 inches (45.72 cm).

3.2.17 Chassis panel assembly. A relay rack mounting panel shall be furnished with each equipment. It shall be constructed as shown in Figure 1 such that one equipment can be fixed on either side of this panel with threaded screws. The panel shall be notched to conform with MIL-STD-189 and shall be complete with one blanking plate. The blanking plate shall be capable of being fixed to either side of the mounting panel with the same threaded screws used to attach the equipment chassis. The two handles on this mounting panel shall be made of corrosion resistant steel in accordance with MIL-E-16400. They shall be placed as shown in Figure 1. In addition to the federal nomenclature plate required on the chassis of each equipment a nomenclature plate shall be fixed on either side of the mounting panel.

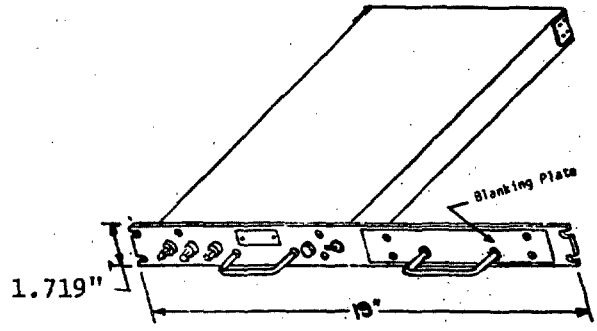


Figure 1. Mounting panel

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3.2.17.1 Chassis panel handling. The design of the chassis with front panel detached, shall be such that it may be placed on a flat surface, on its base, top, or either side without damage to any of its parts.

3.2.17.2 Component mounting. The modular assemblies shall be secured to the main chassis structure with captive screws, and shall not require the use of soldering irons or the disconnection of external leads from terminal strips. The disconnection of coaxial cable fittings will be permitted with reference to this latter requirement. Both types of multicouplers shall have inter-module connections such that the input filter can be bypassed, thereby connecting the input connector directly to the amplifier. Design shall be such that this can be accomplished manually without the use of tools, the removal of circuit components, or the use of additional materials not a part of the equipment. All modules shall use male type BNC connectors for interconnection, with the exception of the power supply and amplifier B+ terminals.

3.2.17.3 Main chassis construction. The main chassis shall be of welded construction. Lapped corners, secured with rivets or screws, will not be acceptable, although rivets or screws will be permitted for brackets, gussets, and the like, where welding would be impractical.

3.2.17.4 Auxiliary shields and cover plates. Auxiliary shields and cover plates employed on the main chassis or modular assemblies, for purposes of electrical shielding, or to provide access to parts contained in modular assemblies, shall be secured with screws which are captive with the removal shield or cover plates. Cover plates shall provide positive electrical contact with their mating bearing surfaces, when secured in place.

3.2.18 Securing devices. Fastening devices shall be in accordance with MIL-E-16400.

3.2.19 Primary power connector. The primary power connector (receptacle) shall be mounted in the rear of the chassis. A mating plug for the connector shall be furnished with the equipment. The connector (receptacle) and its mating plug shall comply with MS 3452 and MS 3456 respectively; the plug shall have an MS 3057 clamp.

3.2.20 Front operating controls and devices. The following controls and devices shall be mounted on the front operating panel. Titles and markings, as indicated, shall be placed on the panel adjacent to their respective items.

3.2.21 Power line switch. The power line switch shall be a toggle switch functioning to turn primary power on and off. It shall be titled POWER with the two positions of the toggle lever marked ON and OFF, with ON applying when the switch lever is in its up position. The switch should open both sides of the primary winding of the power transformer.

3.2.22 Indicator light. An indicator lamp shall be provided to indicate when power is applied to the equipment. The lamp and lamp holder shall be in accordance with MIL-E-16400.

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3.2.23 Circuit design. Circuit design shall be in accordance with the preferred circuits of MIL-E-16400.

3.2.24 Electrical bonding. Electrical bonding shall be in accordance with MIL-E-16400.

3.2.25 Internal cabling. Internal cabling shall be in accordance with MIL-E-16400.

3.2.26 Electrical connections. Electrical connections shall be in accordance with MIL-E-16400.

3.2.27 Wire terminations. Wire terminations shall be in accordance with MIL-E-16400.

3.2.28 Conductor identification. Identification of conductors shall be in accordance with MIL-E-16400.

3.2.29 Safety. Safety design shall be in accordance with MIL-E-16400.

3.2.30 Non-standard parts. Approval of non-standard parts shall be in accordance with MIL-E-16400. Non-standard parts shall not be used in the power supply assembly with the exception of power transformers (see 3.2.6).

3.3 Reliability and maintainability.

3.3.1 Reliability. The equipment shall be designed to achieve a minimum mean-time-between-failure (MTBF) of 31,500 hours for the type I configuration and 34,000 hours for the type II configuration. For demonstration purposes, the specified MTBF (θ_o as defined in MIL-STD-781) shall be 15,000 hours for the type 1 and type II configurations.

3.3.2 Maintainability. The coupler group shall have a mean corrective maintenance time (Mct) not exceeding 30 minutes when repair is accomplished at the organizational level of maintenance by replacement of lowest sub-assemblies (including modules and p.c. boards) and chassis-mounted parts (electronic, electrical, electromechanical and mechanical parts). Repairable subassemblies (modules and p.c. boards) shall have a mean corrective maintenance time (Mct) not exceeding 1.5 hours when repair is accomplished by replacement of individual parts at the appropriate level of maintenance. Corrective maintenance time shall include localization, isolation, disassembly, interchange, reassembly, alignment and checkout for all maintenance tasks.

3.4 Environmental characteristics. The equipment shall have the following environmental characteristics.

3.4.1 Temperature. The multicoupler equipment shall be designed to conform to the temperature requirements for Range 3 temperature of MIL-E-16400. Thermal design shall be in accordance with MIL-E-16400.

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3.4.2 Relative humidity. The equipment shall provide continuous reliable performance in relative humidities ranging up to 95 percent for both continuous and intermittent periods, including conditions wherein condensation takes place in and on the equipment in the form of both water and frost.

3.4.3 Shock. The equipment shall not show evidence of physical damage and shall operate in accordance with the performance requirements specified herein after being subjected to the shock testing specified in 4.4.2.4.

3.4.4 Altitudes. The equipment shall be capable of transportation by aircraft and common carrier, and shall be capable of operating at altitudes up to 10,000 feet (3048 meters) above sea level.

3.5 Electrical performance. The multicoupler shall be designed and constructed to provide the performance specified herein under any possible combination of conditions specified herein. Specific test procedures specified in Section 4 shall apply as pertinent to the particular performance requirement involved.

3.5.1 Antenna input. The antenna input circuit shall be designed for a 50-ohm unbalanced input system.

3.5.2 Number of outputs. The antenna coupler shall provide facilities for the operation of up to eight radio receiving sets from a single antenna (that is eight output connectors).

3.5.3 Output impedance. Each of the eight output connectors shall be designed for operation with an unbalanced coaxial transmission line having a nominal impedance of 50-ohms.

3.5.4 Voltage standing wave ratio (VSWR). The voltage standing wave ratio shall not exceed 1.5:1 to either the input transmission line when terminated with a 50-ohm signal source, or the transmission line for any of the eight outputs when terminated with 50-ohm loads. This requirement shall apply for the input and any given output for any combination of terminations of the other connectors.

3.5.5 Frequency range. The nominal operating frequency range of the complete antenna receiving multicoupler equipment shall be from 2.0 megaHertz (MHz) to 32 MHz with the input filter in the circuit. With the input filter bypassed, the multicoupler shall be able to operate at all frequencies from 0.3 to 2.0 MHz and conform to all other requirements specified herein.

3.5.6 Primary power source. The complete multicoupler shall comply with the performance requirements of this specification when operated from a power source of 105/115/125 volts ±10 percent or 210/230/250 volts ±10 percent, 50 to 60 Hertz ±5 percent, single phase alternating current (AC). It shall also comply with voltage and frequency power transients in accordance with MIL-E-16400.

3.5.6.1 Taps. Taps shall be provided on the primary winding of the power transformer for the voltages specified in 3.5.6. The equipment shall be wired for 115 volts AC. The internal power supply shall supply all necessary voltages

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for proper operation of the multicoupler, when connected to the specified power source.

3.5.7 Electrostatic shielding. The primary winding of the power transformer shall be electrostatically shielded to minimize coupling of radio frequency energy from the power line into the multicoupler circuits. It shall otherwise conform to the requirements of MIL-E-16400.

3.5.7.1 Rectification and regulation. The power supply circuit shall employ solid-state devices for rectification and voltage or current regulation. The regulating circuits shall be designed to prevent any damage to the power supply due to unintentional overloads or short circuits of power supply output. The power supply shall automatically return to normal operation upon removal of the overload or short circuit.

3.5.8 Power input filter. A radio frequency (RF) filter, balanced with respect to ground, shall be provided between the primary power input receptacle and the primary winding of the power supply transformer to attenuate undesired RF signal interference from the power line into the equipment and from the equipment into the power line. The radio frequency filter in the power line shall present an attenuation of not less than 60 decibels (dB) to any radio frequency between the limits of 14 kHz and 40 MHz and 50 dB from 40 MHz to 150 MHz. This requirement is based upon attenuation in either direction, between either line input terminal and any RF output terminal. Fuses as required in 3.2.10 shall be electrically inserted between the primary power input receptacle and the input of the RF filter.

3.5.9 RF input filter. An input filter shall be provided for types I and II multicouplers which gives at least the minimum attenuation from 14 kHz to 150 MHz shown in Figures 2, 3 and 4.

3.5.10 Electromagnetic compatibility (EMC). The antenna coupler shall conform to the requirements of MIL-STD-461 for Class IA equipments: CE01, CE03, CS02, RE02, and RS02

3.5.11 Gain. The insertion gain between the antenna input connector and any output connector with the input filter in the circuit shall be 3.0 ±2 dB throughout the frequency range of 2 to 32 MHz. Insertion gain shall also be 3.0 ±1 dB in the 0.3 to 2.0 MHz frequency range with the input filter bypassed. Both those requirements shall apply with all output connectors terminated in 50 ohms, with all output connectors except the one under test short circuited, and with all output connectors except the one under test open circuited.

3.5.11.1 Gain variation with power line voltage. The signal amplification at any frequency within the specified frequency range shall stay within 3.0 ±2 dB for a change in power line voltage of ±10 percent from the nominal specified voltage.

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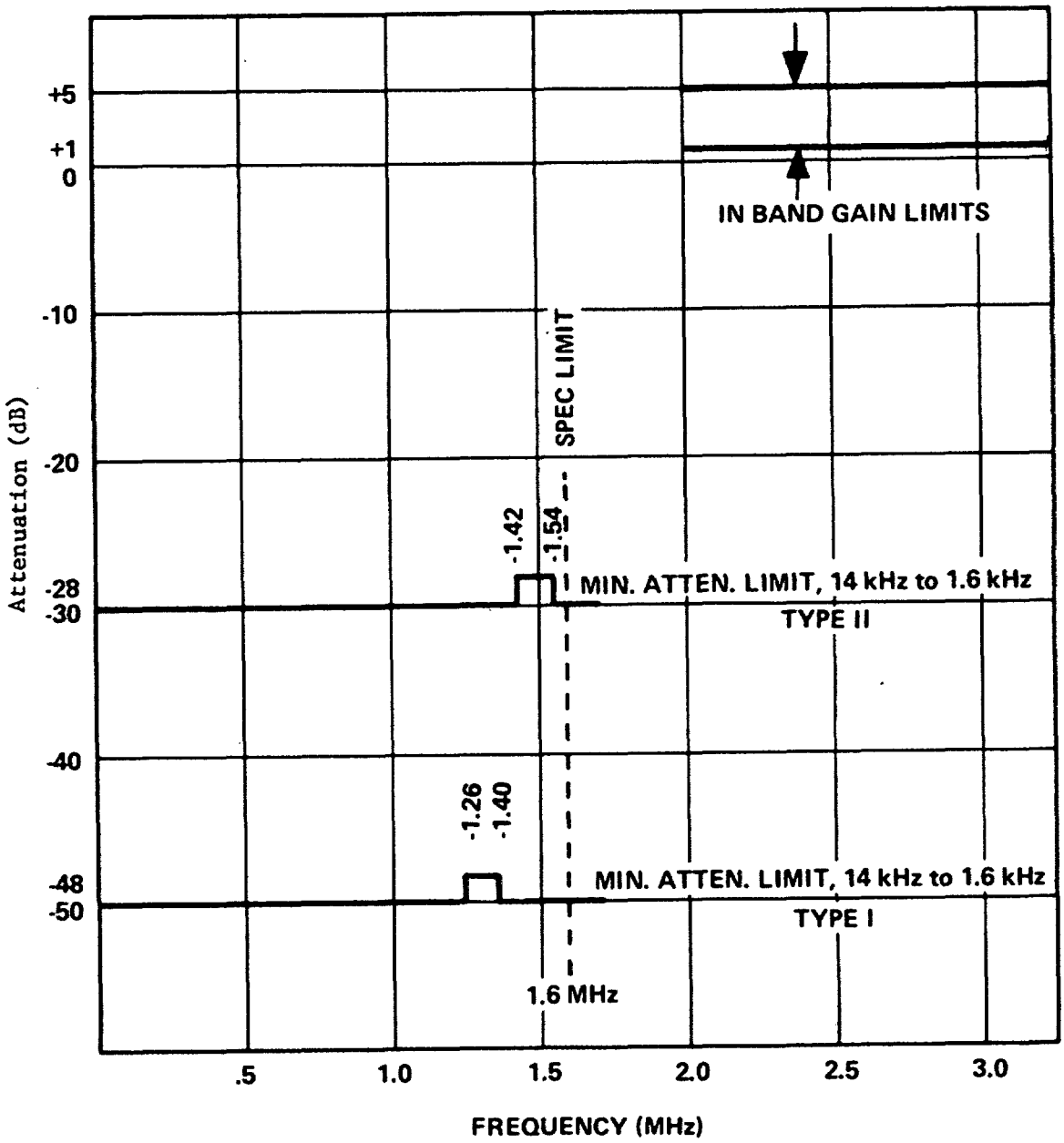


Figure 2. RF filter low frequency

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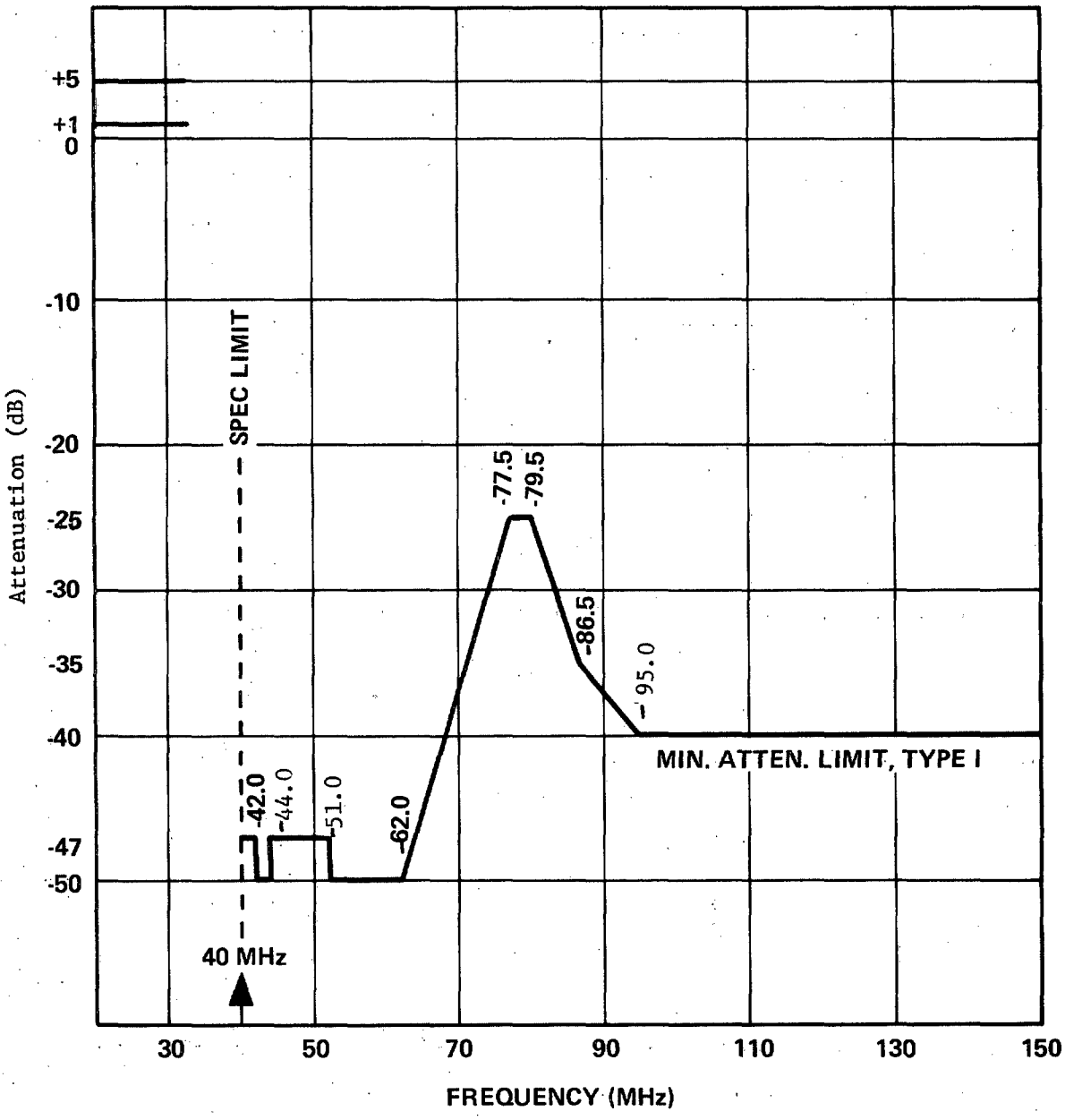


Figure 3. RF filter high frequency

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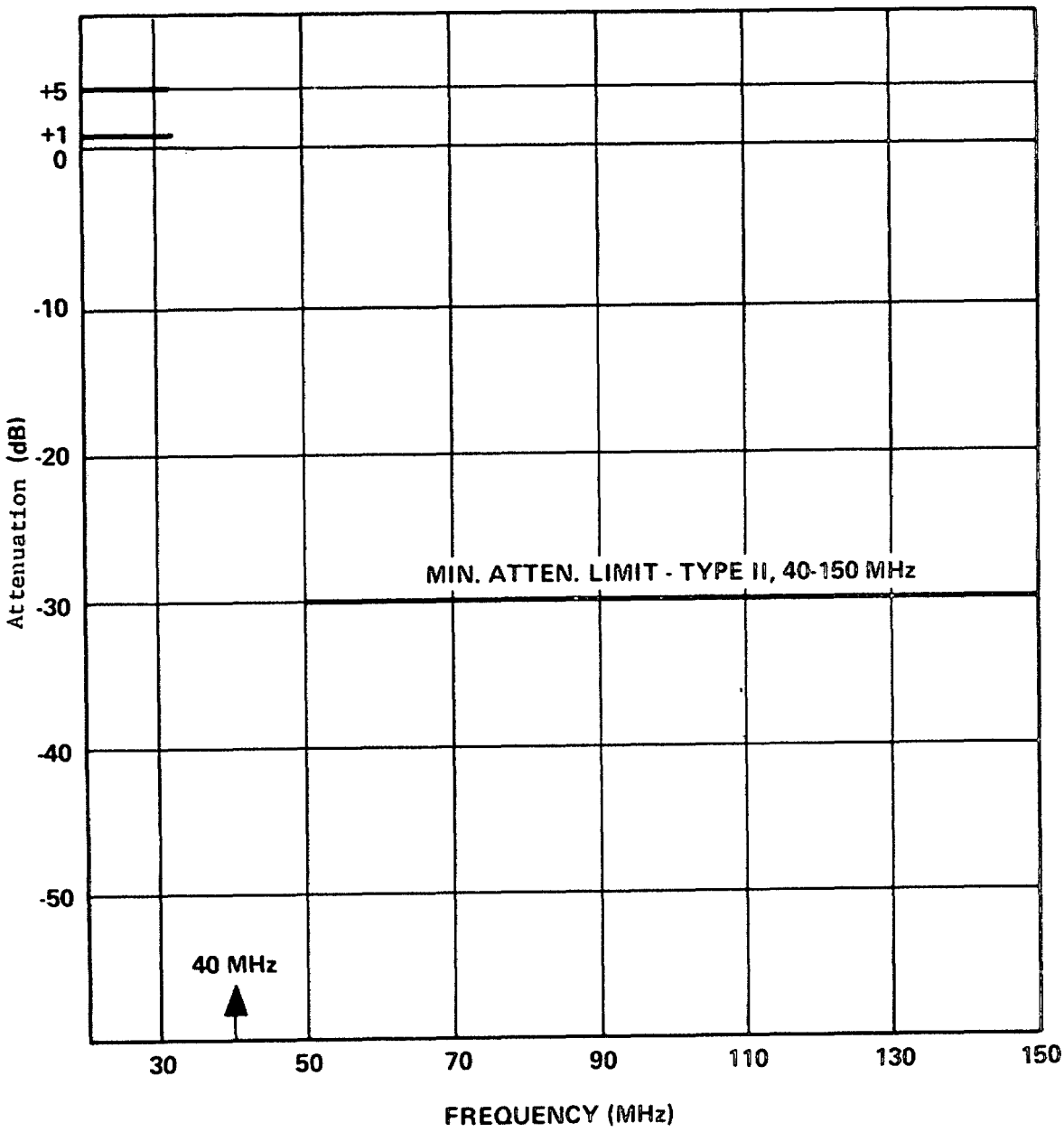


Figure 4. RF filter high frequency

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3.5.11.2 Gain tracking. The type II multicoupler shall have the same gain for all eight outputs of all multicouplers within ±0.5 dB for any given frequency.

3.5.12 Desensitization. The amplification of the desired signal shall not be reduced by more than 3 dB when the level of the undesired signal is increased to 1.5 volts for the Type I multicoupler, and to 1.7 volts for the Type II multicoupler. This shall hold for any undesired signal in the 2 to 32 MHz range with the input filter in the circuit, and any undesired signal in the 0.3 to 32 MHz range with the input filter bypassed.

3.5.13 Overload. The multicoupler shall be designed to allow inputs without causing component failure or subsequent degradation of performance, as follows:

(a) 10 volts rms continuous in the 0.1 to 32 MHz frequency range, with and without RF input filter.
(b) Twenty 500 volt triangular shaped spikes with 5 µsec (microsecond) rise times and 500 µsec fall times, with the spikes not more than 10 minutes apart, with RF input filter connected.

3.5.14 Reradiation (output to input isolation). For the type I and II multicouplers the radio frequency level of any signal present at the antenna connector, as the result of a signal being applied from an external source to any output connector, shall be attenuated 50 dB or more below the RF level of the applied signal.

3.5.15 Output-to-output isolation. The RF level of the signal present at any output connector as the result of a signal being applied from an external source to any other output connector shall be attenuated a minimum of 40 dB or more below the level of the applied signal. This shall hold for all frequencies in the 2 to 32 MHz range with and without the input filter in the circuit. It shall not necessarily hold in the 0.3 to 2.0 MHz frequency range. This isolation shall hold with input connector terminations causing a VSWR as high as 4.0:1.

3.5.16 Spurious products. Spurious products in the 0.3 to 32 MHz frequency range present at the outputs of a multicoupler shall be at least -65 dB relative to 0.5 volts rms when two 0.5 volt rms signals are applied to the multicoupler input. This requirement shall apply with 50-ohm terminations on all unused output ports, with or without the RF input filter connected and for all fundamental frequencies in the 0.3 to 32 MHz range.

3.5.17 Noise figure. The noise figure measured at any frequency within the specified frequency range of the multicoupler shall be less than or equal to 8.0 dB for the Type I multicoupler and less than or equal to 8.5 dB for the Type II multicoupler.

3.5.17.1 Noise figure variation with power line voltage change. The noise figure shall be equal to or better than that specified in 3.5.17 for any power line voltage within ±10 percent of the specified power line voltage.

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3.5.18 Phase correlation. The Type II multicoupler shall have phase correlation between all outputs and all other Type II multicouplers within an absolute range of 2° over the frequency range of 2 to 30 MHz with the RF input filter connected. Additionally, the Type II multicoupler shall have phase correlation with the production phase standard, CU-1382G - Serial Number _____, within ±1° under the same conditions. Figure 5 is representative of the absolute delay versus frequency characteristic of the CU-1382G phase standard. Figure 5 is included for information only, as the phase resolution (20° per division) is not adequate to ensure the required phase correlation.

3.6 Workmanship. Workmanship shall be in accordance with MIL-E-16400.

3.7 Nomenclature plate. Nomenclature plate shall be in accordance with MIL-E-16400.

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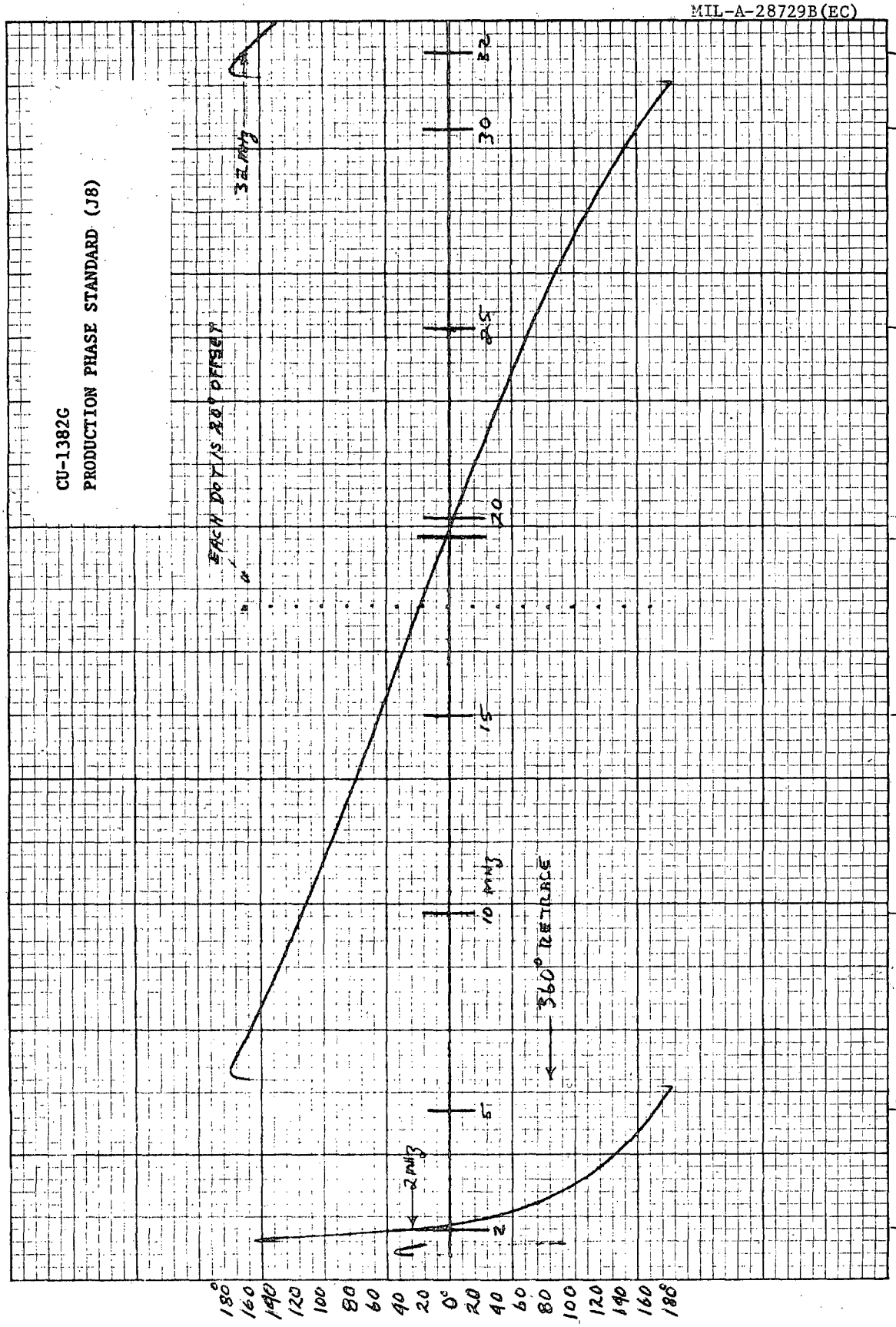


Figure 5. CU-1382G Phase Standard

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will include (a) surveillance of the operations to determine that practices, methods, and procedures are being properly applied, (b) Government product inspection to measure quality of product to be offered for acceptance, and (c) Government product inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.1.2 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

4.2 Classification of inspection. The method of examination and testing shall be classified as follows:

(a) First Article Inspection (4.3)
(b) Quality Conformance Inspection

(1) Production Inspection (4.4.1)
(2) Environmental Inspection (4.4.2)

(c) Reliability Testing (4.5)
(d) Maintainability Demonstration (4.6)
(e) Inspection of Preparation for Delivery (4.7)

4.2.1 Test conditions. Unless otherwise specified, tests shall be performed at factory ambient temperature, humidity, and pressure. Input power shall be 115 volts ±10 percent at 60 Hz ±5 percent.

4.3 First article inspection. First article inspection shall consist of all examinations and testing necessary to determine compliance with the requirement of this specification. First article inspection shall include the examination and testing specified in Table I. Unless otherwise specified one unit shall be required for first article inspection (see 6.3).

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Table I. First article inspection.

Inspection or test	Requirement	Reference
Voltage standing wave ratio	3.5.4	4.4.1.1
Gain	3.5.11	4.4.1.2
Gain tracking	3.5.11.2	4.4.1.3
Desensitization	3.5.12	4.4.1.4
Output-to-output isolation	3.5.15	4.4.1.5
Noise figure	3.5.17	4.4.1.6
Absolute phase delay	3.5.18	4.4.1.7
Primary power source	3.5.6	4.4.1.15
Noise figure variation	3.5.17.1	4.4.1.6
Overload	3.5.13	4.4.1.9
Reradiation	3.5.14	4.4.1.10
Power input filter	3.5.8	4.4.1.11
RF input filter	3.5.9	4.4.1.12
Spurious products	3.5.16	4.4.1.13
DC power supply	3.2.2	4.4.1.14
Radio frequency interference	3.5.10	4.4.2.1
Temperature	3.4.1	4.4.2.2
Relative humidity	3.4.2	4.4.2.3
Shock	3.4.3	4.4.2.4
Altitude	3.4.4	4.4.2.5

4.4 Quality conformance inspection.

4.4.1 Production inspection. Production inspection, including sampling, shall conform to Table II and to the inspection procedures of MIL-STD-105 using the general inspection levels.

Table II. Production inspection.

Inspection or test	Requirement	Reference	AQL percent
VSWR	3.5.4	4.4.1.1	2.5
Gain	3.5.11	4.4.1.2	2.5
Gain tracking	3.5.11.2	4.4.1.3	2.5
Desensitization	3.5.12	4.4.1.4	1.5
Output-to-output isolation	3.5.15	4.4.1.5	1.5
Noise figure	3.5.17	4.4.1.6	1.5
Phase correlation	3.5.18	4.4.1.8	1.0
Primary power source	3.5.6	4.4.1.1.6	2.5
Noise figure variation with line voltage change	3.5.17.1	4.4.1.6	2.5
Overload	3.5.13	4.4.1.9	1.5
Reradiation	3.5.14	4.4.1.10	2.5
Power output filter	3.5.8	4.4.1.11	2.5

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Table II. Production inspection (continued).

Inspection or test	Requirement	Reference	AQL percent
RF input filter	3.5.9	4.4.1.12	1.0
Spurious products	3.5.16	4.4.1.13	1.0
DC power supply	3.2.2	4.4.1.14	2.5

4.4.1.1 VSWR. Voltage standing wave ratio shall be measured using sweep frequency techniques in the 2 to 32 MHz range with the input filter in the circuit. This shall be done for the input and all eight outputs. In addition, the same measurements shall be made in the 0.3 to 2.0 MHz range with the input filter bypassed.

4.4.1.2 Gain. Gain specified in 3.5.11 shall be determined by making sweep frequency gain measurements throughout the 2 to 32 MHz range with the input filter in the circuit. In addition, sweep frequency gain measurements shall be made in the 0.3 to 2 MHz range with the input filters bypassed. Both gain measurements shall be made and results reported for each output circuit under each of the following conditions:

(a) With all outputs except the one under test open circuited.
(b) With all outputs except the one under test short circuited.
(c) With all outputs terminated in independent 50 ohm loads.

4.4.1.2.1 Gain measurements. A 50-ohm noninductive resistor shall be connected as a dummy load on the output connector of the output channel being measured and the output voltage across this resistor measured with a vacuum tube voltmeter. An unmodulated signal shall be applied to the antenna connector from a 50 ohm signal source. The voltage measured at the input connector shall be measured with a vacuum tube voltmeter. Care shall be taken to preclude the test input voltage being a level high enough to overload one of the circuits. The gain shall be the ratio of the output voltage to the input voltage.

4.4.1.3 Gain tracking. In addition to the measurements specified in 4.4.1.2, the type II multicoupler shall have each output measured to determine conformance with 3.5.11.2.

4.4.1.4 Desensitization. Measurements shall be made with the input filter in the circuit to determine conformance with 3.5.12 with a desired signal having an input level equivalent of 100 microvolts and any convenient frequency, between the limits of 2.0 to 32 MHz, also with its output level measured at one output circuit. The undesired signal may have any convenient frequency between 2.0 and 32 MHz. In addition, the input filter shall be bypassed and a desired signal of 100 microvolts at 0.3 MHz and an undesired signal of 0.5 MHz shall be used to make the same measurement. Desired signal gain measurements shall be made for each 10 dB of increase of level of the undesired signal from an initial input level equivalent to 250 microvolts. The results of the measurements shall be plotted on cross section paper to show a curve of desired signal gain with respect to undesired signal level.

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4.4.1.4.1 Desensitization measurement. The purpose of this measurement is to determine the radio frequency voltage level applied to the antenna coupler input connector, which will cause a 3 dB reduction in overall antenna coupler gain offered to a comparatively weak signal on a different frequency.

4.4.1.4.2 Two radio frequency voltage sources are required for this test. One voltage source shall be a conventional signal generator. The other voltage source shall be capable of providing several volts of radio frequency energy to the multicoupler from a 50 ohm source.

4.4.1.4.3 The signal generator shall be connected to the multicoupler input through a decoupling network. The high voltage source shall also be coupled to the multicoupler input thru this decoupling network. The two signal generators and the decoupling network shall be such that their interaction results in a 50 ohm output impedance.

4.4.1.4.4 The measuring receiver or vacuum tube voltmeter shall be connected directly to one of the antenna coupler outputs.

4.4.1.4.5 The signal generator shall be adjusted to produce 100 microvolts and the receiver shall be adjusted to read a convenient level with no voltage from the other source.

4.4.1.4.6 The source of strong signals shall be adjusted to a frequency far enough removed from the signal generator frequency to preclude cross-modulation and overload in the measuring receiver.

4.4.1.4.7 The interfering voltage from the strong source shall be measured at the antenna coupler input connector by means of vacuum tube voltmeter or a decoupled voltmeter receiver.

4.4.1.4.8 The signal generator voltage shall be adjusted, for incremental 10 dB increases in interfering signal, in order to maintain constant output from the measuring receiver.

4.4.1.4.9 The desensitization voltage shall be the interfering signal voltage, measured at the antenna coupler input, which causes the coupler gain to decrease by 3 dB.

4.4.1.5 Output-to-output isolation. Measurements to verify compliance with 3.5.15 shall be made using sweep frequency techniques in the 2 to 32 MHz range. These measurements shall be made with the input filter in the circuit for input connector terminations of 50, 12-½, and 200 ohms. For each input connector termination, measurements shall be made under the following conditions:

(a) With all outputs except the two under test open circuited.
(b) With all outputs except the two under test short circuited.
(c) With all outputs terminated in independent loads of 50 ohms.

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4.4.1.6 Noise figure. Noise figure measurements to verify compliance with 3.5.17 shall be made at frequencies of approximately 2.0, 3.0, 4.0, 6.0, 8.0, 12.0, 16.0, 20.0, 24.0, 28.0, and 32.0 MHz for any one output circuit with the input filter in the circuit. Additional measurement shall be made at 0.3, 0.7, 1.0, and 1.5 MHz with the input filter bypassed. All of the above measurements shall be made at nominal operating voltage 10 percent below nominal operating voltage, and 10 percent above it. The above measurements shall be made with all outputs terminated in independent 50 ohm loads.

4.4.1.6.1 Noise figure measurement. The noise figure measurement shall be made using a commercial noise generator, noise diode, or equivalent, and with a receiver having a sensitivity of at least 2 microvolts for a signal plus noise to noise ratio of 20 dB and a minimum bandwidth of 10 kHz. The radio receiver is required as a detector. The noise figure measuring equipment shall be suitable for use over the specified frequency range and with a 50-ohm radio frequency system. The multicoupler-receiver combination shall be considered as two networks in series and the noise figure of the antenna coupler computed from:

$$F_a = F_{ab} - \frac{F_b - 1}{G_a}$$

or in dB:

$$F_a \text{ (dB)} = 10 \log_{10} \left[F_{ab} - \frac{F_b - 1}{G_a} \right]$$

F_{ab} = Noise figure of the combination in dB.
F_a = Noise figure of multicoupler expressed as a power ratio.
F_b = Noise figure of the receiver expressed as a power ratio.
G_a = Available power gain of the multicoupler with the input filter in the circuit.

4.4.1.7 Absolute phase delay. Overall phase shift measurement for the type II multicoupler over the full specified operating frequency range, with RF input filter connected, of 2 to 32 MHz shall be made in accordance with the following procedure to determine compliance with 3.5.18. NOTE: This test shall apply only to first article inspection.

4.4.1.7.1 Using the test setup shown in figure 6 with the calibration phase delay cable in place, set the sweep generator output to -10 dbm in the 1 to 32 MHz range for an initial setup condition. (Note: Test equipment and device under test shall be at thermal equilibrium; a minimum of four hours on-time for all devices shall be accumulated prior to test.)

4.4.1.7.2 Connect cables AB and AC together. Operate the network analyzer in a phase detection mode with 1° per centimeter scale factor. Determine that display trace is within ±0.2 cm of the center line. Adjust cable lengths if trace is not within ±0.2 cm across the band of interest. Return to this step and adjust the phase vernier knob anytime test equipment drift (vertical) is suspected.

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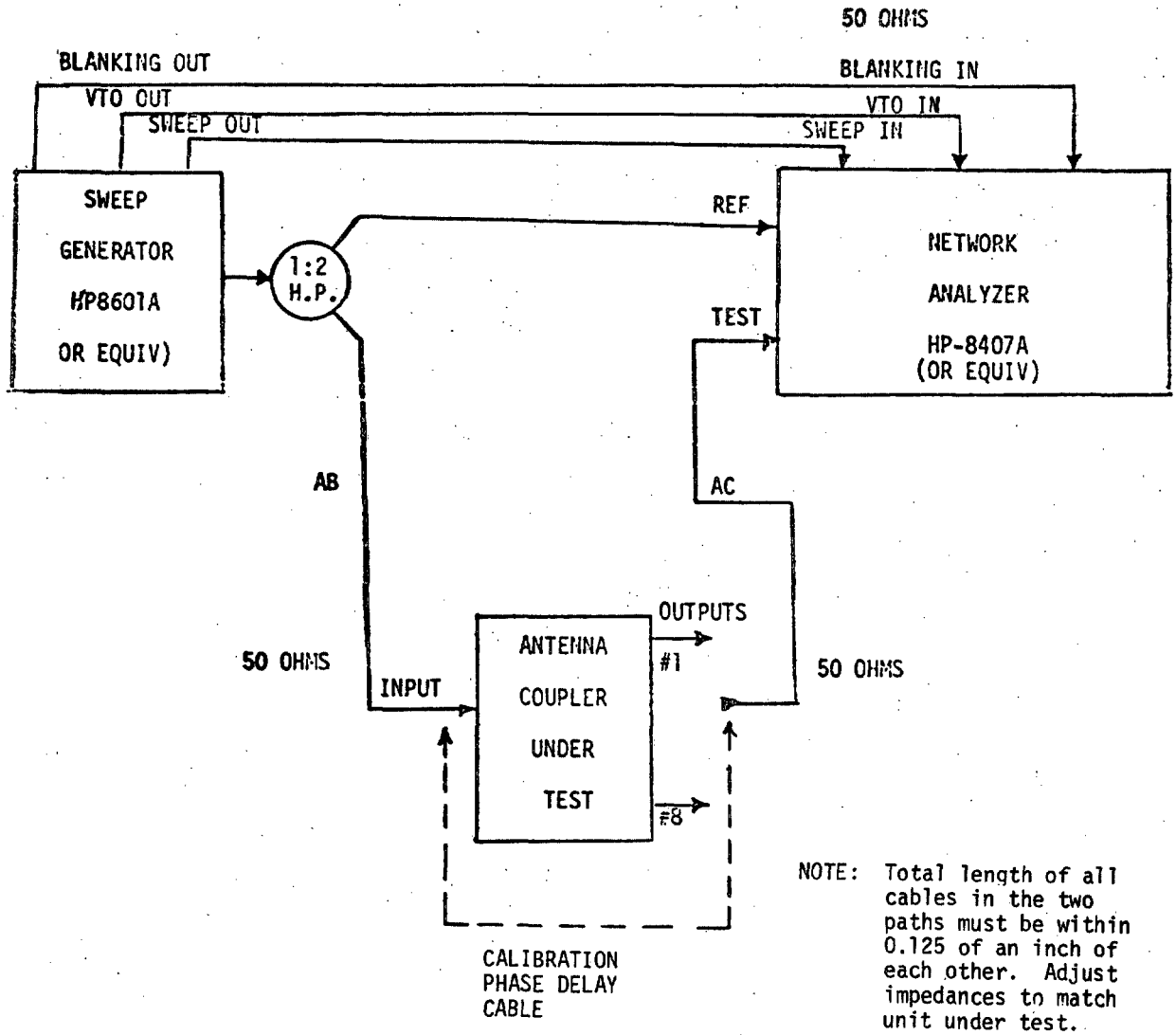


Figure 6. Test setup - absolute phase measurement

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4.4.1.7.3 Insert the calibration phase delay cable in the test channel. In the 0° offset position with the offset switch set to (-), switch 8601A sweep generator to the CW mode, adjust the frequency until the beam is on the center line of the scope. Do not record the retrace crossing when the sweep goes from 360° to 0°. The retrace crossing will show up as a wide vertical line, not a dot that can be centered. Record four consecutive frequencies (at zero crossing points) on the calibration data sheet (figure 7).

Measured frequency	The three different frequencies should be the same (6 kHz max error)
f1 =	XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
f2 =	f1 - f2 = f5 _____
f3 =	f2 - f3 = f6 _____
f4 =	f3 - f4 = f7 _____

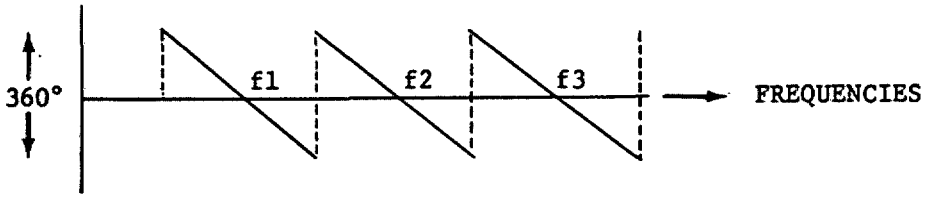


Figure 7. Measured frequencies

4.4.1.7.4 Frequencies.

(a) Average the difference frequency (f5, f6, f7) f8 = _____

(b) Divide f8 by 360° to obtain frequency in KC per degree. f9 = ____

(c) Multiply f9 by 20° = _____ f9 x 120° = _____
f9 x 40° = _____ f9 x 140° = _____
f9 x 60° = _____ f9 x 160° = _____
f9 x 80° = _____ f9 x 180° = _____
f9 x 100° = _____ (These are the true (expected) frequencies.)

(d) Add these products to f1 when the offset switch is in the (-) position. These are the true frequencies that the offset should shift the frequency. Record the sums on the (-) degree side of the true frequency column (Figure 9).

(e) Subtract these products from f1 when the offset switch is in the (+) position. Record the difference on (+) degree side of the true frequency column (Figure 9).

Figure 8. Frequencies (f9)

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8407A Nominal offset	True frequency (calculated)	Correction factor for 8407A offset (+ or - degrees, measured)	Include 0°/ offset from step 2
+180°			
+160°			
+140°			
+120°			
+100°			
+ 80°			
+ 60°			
+ 40°			
+ 20°			
0° (f1)			
- 20°			
- 40°			
- 60°			
- 80°			
-100°			
-120°			
-140°			
-160°			
-180°			

8407A Serial # _____

Date _____ Engr._____

Figure 9. 8407A Calibration data

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4.4.1.7.5 Return to the first frequency that was recorded (f1). Insert 0° offset on the front panel of the 8407A. The beam should be again on the center line of the scope. If it is not, return to 4.4.1.7.2. Adjust the 8601 to the true frequency recorded in the -20° column, Figure 8. A precise adjustment must be made by the use of a frequency counter. Insert -20° offset, the scope dot will appear very close to the center line. The -20° offset error is the displacement of the beam from the center line. The phase degree per division switch is still set on one degree per division, so the error can now be recorded to the nearest 1° in the column provided on Figure 8. If the scope beam is above the center line the error is recorded as (+) degrees. Repeat this procedure for phase offsets -40° to -180°. Next, switch the phase offset switch to (+), and repeat this (these) step(s) for +20° to +180°. Record each error in the appropriate column. Also include zero offset (±0) seen in 4.4.1.7.2 due to cable and equipment phase tracking errors at each calculated true frequency (±Ø offset). Add this zero Ø error to other calibration data.

4.4.1.7.6 Replace the calibration phase delay cable with the multicoupler under test, and connect cable AB to the multicoupler input and cable AC to output #1. With the 8601A sweep generator still in the CW mode insert 0° phase offset on the front panel of the 8407A and measure accurately with the electronic counter, the frequencies at which the multicoupler phase shift is 0°. There will be at least two between 2 and 32 mc. Record on data sheet, Figure 10.

4.4.1.7.7 Insert -20° offset on the 8407 front panel, and repeat 4.4.1.7.6. Instead of using the center line to zero the scope beam, offset the scope dot to the correction factor (in degrees) obtained in Figure 8 for the -20° offset include step 2 zero offset. Record the frequency in the -20° column on the Table of Figure 9.

4.4.1.7.8 Repeat 4.4.1.7.7 for each offset (-40° to -180° and +20° to +180°). For each offset the scope beam must be displaced from the center line by the correction factor for that offset. The data is complete when frequency readings are obtained across the 2 to 32 mc frequency range in 20° steps. With the filter in place, this multicoupler has about 540° of phase shift. More than one column (350°) of data will be collected.

4.4.1.8 Phase correlation. The relative phase outputs of each receiving multicoupler and between each multicoupler shall be measured in accordance with the following procedure to determine compliance with 3.5.18.

Initial test conditions:

(a) RF filter connected
(b) Unused ports terminated in 50 ohms.

NOTE: All multicouplers and test equipment used shall have been turned on and operating normally four hours prior to commencement of calibration of test equipment and subsequent phase measurements.

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Nominal phase shift from 8407A dia	Measured frequency data column #1	Measured frequency data column #2
+180°		
+160°		
+140°		
+120°		
+100°		
+ 80°		
+ 60°		
+ 40°		
+ 20°		
0°		
- 20°		
- 40°		
- 60°		
- 80°		
-100°		
-120°		
-140°		
-160°		
-180°		

8407A Serial #_____ Multicoupler serial # _____

Date _____ Engr. _____

Figure 10. Multicoupler phase shift

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4.4.1.8.1 Configure the test setup as shown in Figure 11. Connect cables AB to AC, AD to AE.

4.4.1.8.2 Set the sweep generator for a nominal output of -10 dBm in the 1 to 32 MHz range. Operate the network analyzer in a phase detection mode with a 1°/centimeter scale factor; verify that the display trace is within ±0.2 cm of the center line. Adjust cable lengths if trace is not within ±0.2 cm across the band of interest. Return to this step and adjust the phase vernier knob whenever test equipment drift is suspected.

4.4.1.8.3 Connect the phase standard multicoupler and the test multicoupler into the test setup as indicated in Figure 11. (Connect AE to phase standard multicoupler input, AD to output 1; connect AB to test multicoupler input, AC to output #1.) Observe phase difference in the 2 to 32 MHz range with a 1° per centimeter deflection factor.

4.4.1.9 Overload. RF input overload tests shall be performed by first applying 10 volts rms at 32 MHz to this input connector for a continuous period of 10 minutes. Next, 20 voltage spikes as described in 3.5.13 shall be applied to the input connector with the voltage spikes not greater than 10 minutes apart. The overload tests shall be conducted with and without the RF input filter connected for CW overload (10 VRMS at 32 MHz) and only with the RF input filter connected for voltage transient overload.

4.4.1.10 Reradiation. Measurements to verify compliance with 3.5.14 shall be made using sweep frequency techniques in the 2 to 32 MHz range with the input filters in the circuit. In addition, this measurement shall be made in the 0.3 to 2.0 MHz range with the input filter bypassed. Both of the above measurements shall be made with the signal applied consecutively to each output connector.

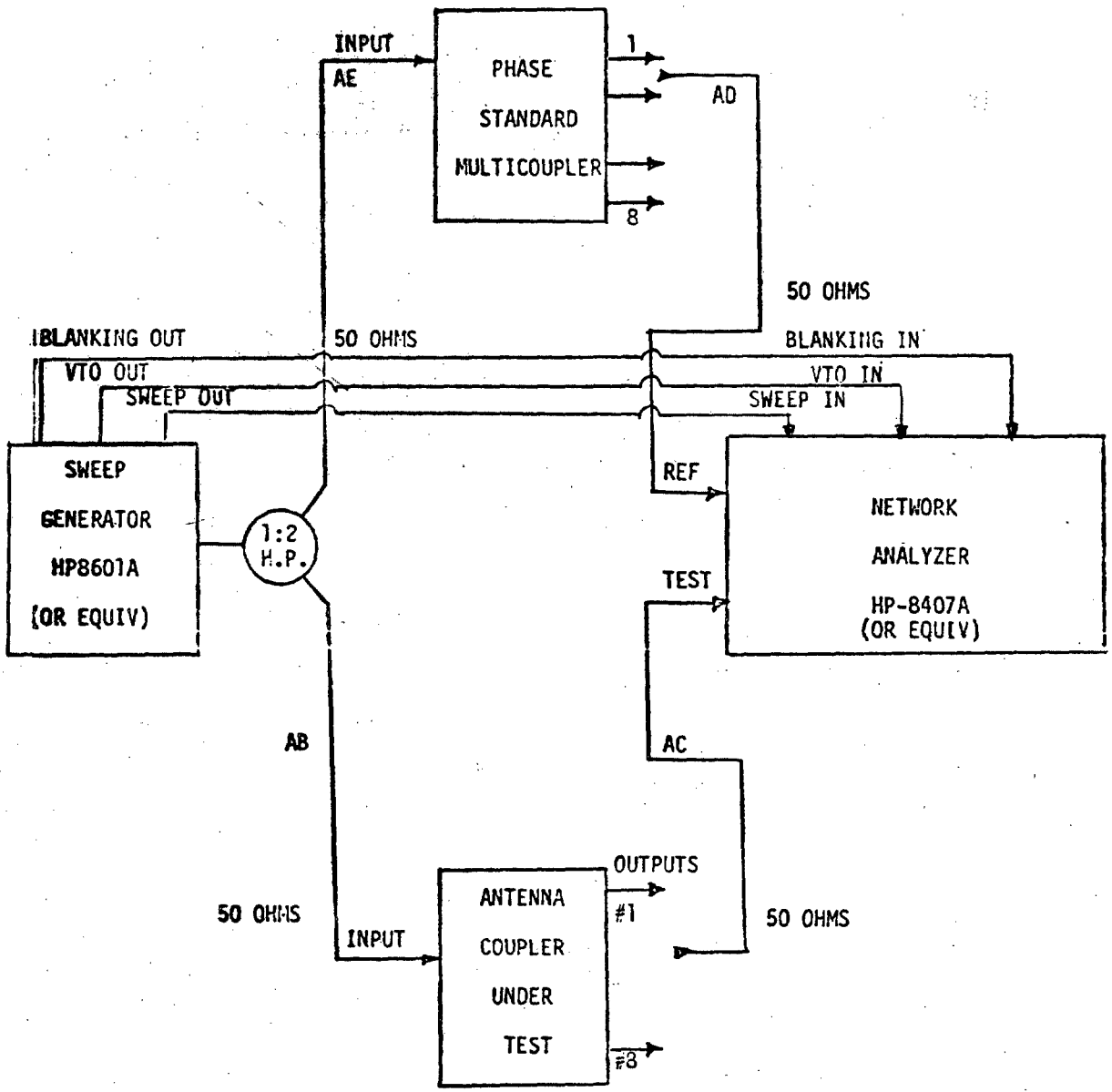
4.4.1.11 Power line filter. Compliance with 3.5.10 shall be determined by measuring the gain of the amplifier at any output connector with the signal coupled to the power line and with the antenna connector terminated and shielded so as not to contribute signal to the output.

4.4.1.11.1 Power line filter frequency attenuation measurement. Attenuation of signal shall be measured by superimposing a signal level at any output. The antenna input connector shall be terminated with a 50-ohm resistance and shall be shielded to prevent any signal entrance through the normal input circuit.

4.4.1.12 RF input filter. RF input filter attenuation shall be measured by sweeping the frequencies from 14 kHz to 2.0 MHz and 32 MHz to 150 MHz, and measuring the attenuation (see 3.5.9).

4.4.1.12.1 Input filter measurement. Measurements shall be made to determine agreement with 3.5.9 by applying the test signal to the antenna connector and measuring the signal level across the eight RF output parts.

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NOTE: Total length of all cables in the two paths must be within 0.125 of an inch of each other. Adjust impedances to match unit under test.

Figure 11. Test setup - relative phase measurement

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Output No.	Phase difference 2 MHz	8 MHz	16 MHz	32 MHz
1	____	____	____	____
2	____	____	____	____
3	____	____	____	____
4	____	____	____	____
5	____	____	____	____
6	____	____	____	____
7	____	____	____	____
8	____	____	____	____

Test multicoupler serial # _____

Figure 12. Relative phase measurement

4.4.1.13 Spurious products. Measurements to determine conformance with 3.5.16 shall be made at the following frequencies. All measurements shall be made with the input filter in the circuit, unless otherwise noted.

f_1 (MHz)	f_2 (MHz)
2.1	3.0
3.5	5.0
5.6	8.0
7.7	11.0
11.2	16.0
15.4	22.0
21.0	30.0
3.5	3.0
2.2	3.0
15.4	22.0
0.7 [1/]	0.3 [1/]
0.8 [1/]	0.5 [1/]

[1/] To be performed with input filter bypassed.

4.4.1.13.1 Spurious products measurement. Spurious products shall be measured in a test system with residual spurious products less than 90 dB below two 0.5 VRMS signals relative to 0.5 VRMS to determine compliance with 3.5.16. All in-band spurious products shall be measured simultaneously for each pair of fundamental frequencies in 4.3.4. All unused RF ports shall be terminated in 50 ohms.

4.4.1.14 DC power supply. During normal operation of the multicouplers, the power supply leads shall be shorted five times for 30 seconds each time (see 3.5.7.1).

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4.4.1.15 Primary power source. Steady state voltage and frequency, transient voltage and transient frequency shall be performed at the voltages and frequencies specified in 3.5.6. The test method shall be as described in MIL-E-16400. Gain shall be measured under these conditions to determine conformance with 3.5.11.

4.4.2 Environmental inspection. Environmental inspection shall be as listed in Table III and shall be performed on units which have been subjected to and passed production inspection.

Table III. Environmental inspection.

Inspection or test	Requirement	Reference
Radio frequency interference	3.5.10	4.4.2.1
Temperature	3.4.1	4.4.2.2
Relative humidity	3.4.2	4.4.2.3
Shock	3.4.3	4.4.2.4
Altitude	3.4.4	4.4.2.5

4.4.2.1 Electromagnetic compatibility test. The requirements of 3.5.10 shall be verified by tests performed in accordance with MIL-STD-462.

4.4.2.2 Temperature tests. The temperature test shall be that specified in MIL-E-16400 for range 3 temperature equipment. During the temperature operating test, gain measurements shall be made at each output connector with all outputs except the one under test terminated. These measurements shall be made at the lowest and highest of the specified temperature limits and at room temperature after completion of the test. Each measurement shall be made at 0.3, 12, 17, and 32 MHz with the input filter bypassed.

4.4.2.3 Relative humidity. The humidity test shall be performed in accordance with the test methods of MIL-E-16400 (see 3.4.2).

4.4.2.4 Shock. The packaged test item shall be dropped cornerwise onto a hard level concrete floor or equal surface on each of its eight corners, falling freely through the vertical distance of 30 inches (762 mm). Prior to each drop, the package shall be suspended with its center of gravity vertically above the striking corner. After completion of all drops, the equipment shall be unpacked and subjected to the gain test in 3.5.11.

4.4.2.5 Altitude. The equipment shall be subjected to the altitude test, Method 500, Procedure I, of MIL-STD-810. Gain measurement of 4.4.1.2 and noise figure of 4.4.1.6 shall be measured at 10,000 feet (3048 meters) and compared to the production test data to determine degradation. The same measurements shall be taken upon completion of tests.

4.4.3 Sampling for environmental inspection. One equipment shall be selected from each successive 200 equipments, or fraction thereof, produced. The first sample equipment shall be selected from the first months production.

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4.5 Reliability demonstration test. The reliability of the equipment shall be demonstrated in accordance with test plan XXV of MIL-STD-781. Temperature cycling between -40°C and +50°C shall be performed throughout the test in accordance with the Temperature Cycling paragrapn of MIL-STD-781. Vibration, input voltage cycling and on-off cycling are not required. For purposes of the demonstration, each equipment type shall have a specified MTBF (θo)of 15,000 hours.

4.5.1 Test sample. The sample size shall be in accordance with MIL-STD-781. When a contract involves both Type I and Type II, the test sample shall include an equal number of each type.

4.5.2 Rejected equipment. In the event equipment is rejected as a result of the reliability demonstration, the provisions of the Reject in Reliability Qualification (Demonstration) Phase paragraph of MIL-STD-781, shall apply.

4.6 Maintainability demonstration. A maintainability demonstration of coupler repair by replacement of faulty subassemblies (modules and p.c. boards) and chassis-mounted parts shall be performed in accordance with MIL-STD-471. The demonstration shall consist of 20 corrective maintenance actions involving faults selected at random from a list of 40 candidate faults. The candidate faults shall be determined in accordance with Appendix A of MIL-STD-471. Faults shall be created by the introduction of faulty parts or by the simulation thereof.

4.6.1 Technicians. The Command or Agency concerned reserves the right to provide and select the technicians to perform the maintainability demonstration. The contractor shall provide adequate training to qualify the military technician(s) to perform the maintainability demonstration.

4.6.2 Technical documentation Technical documentation to be used in the demonstration shall be limited to the technical manual.

4.6.3 Calculation of demonstrated M_{ct}. The demonstrated M_{ct} shall be calculated by averaging the times recorded for the 20 corrective maintenance actions.

4.6.4 Acceptance/rejection. The demonstrated M_{ct} shall be compared to the specified M_{ct} to determine acceptance/rejection. If a reject decision is reached, the Command or Agency concerned shall be immediately notified. The contractor shall, at no additional cost to the Government:

(a) Develop an approach for redesign or correction of all deficiencies.
(b) Upon approval of a documented approach, the contractor shall repeat the demonstration until an accept decision is reached.

4.7 Inspection of preparation for delivery. Inspections shall be conducted to ensure conformance with the requirements of Section 5.

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5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation and packaging, packing, and marking. Unless otherwise specified, preparation for delivery shall be in accordance with MIL-E-17555 (see 6.2).

6. NOTES

6.1 Intended use. This equipment is intended for use ashore.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Number of first article samples to be submitted, if other than as specified in 4.3.
(c) Type of equipment (see 1.2).
(d) Marking, levels of preservation and packaging and packing required (see 5.1).

6.3 First article. Invitations for bids should provide that the Government reserves the right to waive the requirement for first article samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.4 Environmental inspection. Approval to ship may be withheld at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action.

6.4.1 Initial shipments. Initial shipment of end items will be contingent upon successful completion of the environmental inspection.

6.5 CHANGES FROM PREVIOUS ISSUE. THE OUTSIDE MARGINS OF THIS DOCUMENT HAVE BEEN MARKED "#" TO INDICATE WHERE CHANGES (DELETIONS, ADDITIONS, AND SO FORTH) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS HAS BEEN DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT AS WRITTEN IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

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